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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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SPELL-CHECKING APPLICATION – CHOOSING THE BEST DATA STRUCTURE

CSA0310-DATA STRUCUTURE

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INTRODUCTION

- A **spell-checking application** checks whether a typed word is present in a dictionary and identifies spelling errors.
- Since the dictionary may contain thousands or millions of words, fast searching is essential.
- A **Trie (Prefix Tree)** is the most suitable data structure because it stores words character by character, allowing quick word lookup and prefix searching.
- It also supports features such as **autocomplete** and **spell correction**, making it an efficient choice for modern spell-checking systems.

PROBLEM STATEMENT

Problem

A spell-checking application needs to verify whether a typed word exists in its dictionary database.

Requirements

- Fast word lookup
- Efficient handling of large dictionaries
- Quick suggestions for incorrect words
- Low search time

SUITABLE DATA STRUCTURE – TRIE (PREFIX TREE)

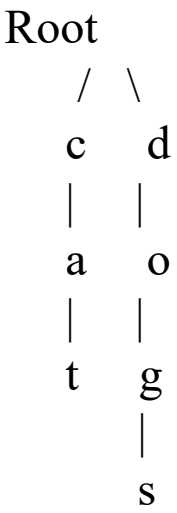
Definition

A **Trie** is a tree-based data structure used to store and search strings efficiently.

Features

Each node represents one character.
Words are formed from the root to leaf.
Supports prefix-based searching.
Ideal for dictionaries and autocomplete systems.

WORKING OF TRIE



Stored Words:

cat

dog

dogs

Search Example:

Search "cat" → Found ✅

Search "cow" → Not Found ❌

JUSTIFICATION BASED ON SEARCH SPEED
WHY TRIE?

Searches character by character.
No need to compare the entire dictionary.
Lookup time is **O(L)** regardless of the number of words.
Performance remains fast even with millions of words.

Example
Dictionary:
apple
application
apply
apt
Searching **"apply“**

Traverse only:
a → p → p → l → y
Search completes in **5 steps**, independent of dictionary size.

WHY TRIE IS THE BEST CHOICE?

- Very fast search
- Search depends only on word length
- Supports autocomplete
- Easy spell correction
- Efficient prefix matching
- Widely used in search engines and dictionaries

Applications

Trie is used in:

•Spell-checking software
•Autocomplete keyboards
•Search engines
•Dictionary applications
•Text prediction
•Contact search

CONCLUSION

- A **Trie (Prefix Tree)** is the most suitable data structure for a spell-checking application because it provides **fast and efficient word searching**.
- Its search time depends only on the **length of the word**, making it ideal for large dictionaries.
- In addition to quick word verification, it supports **autocomplete**, **prefix search**, and **spell correction**, improving both performance and user experience.
- Therefore, a Trie is the preferred choice for modern spell-checking systems.

THANK YOU